

LAB 3: Propositional Logic & Model Checking

Group Challenge

Instructor	Dr. Naila Marir	Duration	1 hour 30 minutes
Semester	Fall 2026	Group size	3-4 students
Submission	One GitHub link per group	Total marks	100

Group No.			
Member name	Student ID	Member name	Student ID
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1. Objectives

1. Represent facts as propositional logic sentences using Not, And, Or and Implication.
2. Explain how `model_check` decides whether a knowledge base entails a query.
3. Encode a new logic puzzle in Python and solve it with model checking.

2. Tools and Files

1. Python 3 and the package `termcolor` (install with `pip install termcolor`).
2. Lab files (on Blackboard): `logic.py`, `harry.py`, `clue.py`, `puzzle.py`, `mastermind.py`. Put all of them in one folder.
3. A GitHub account for each group.

3. Lab Plan

Step	Activity	Time
0	Setup: run <code>harry.py</code> and <code>clue.py</code>	5 min
1	Task 1 – Understand the code	20 min
2	Task 2 – Change the knowledge	20 min
3	Task 3 – Knights & Knaves challenge	30 min
4	Write the PDF and push to GitHub	15 min

4. Task 1 – Understand the Code (30 marks, 20 min)

1. `harry.py`: Write the truth table for `rain`, `hagrid`, `dumbledore` (8 rows). Mark the rows where the knowledge base is true. Use the table to explain why the program prints `True`.

Task 1: harry.py

(1)

Knowledge: ~~Not~~ Not rain $\rightarrow$ hagrid
~~Not rain~~ And hagrid $\vee$ dumbledore
 And Not hagrid $\wedge$ ~~Not dumbledore~~

Rain	hagrid	dumbledore
True	True	True
True	True	False
True	False	True
True	False	False
False	True	True
False	True	False
False	False	True
False	False	False

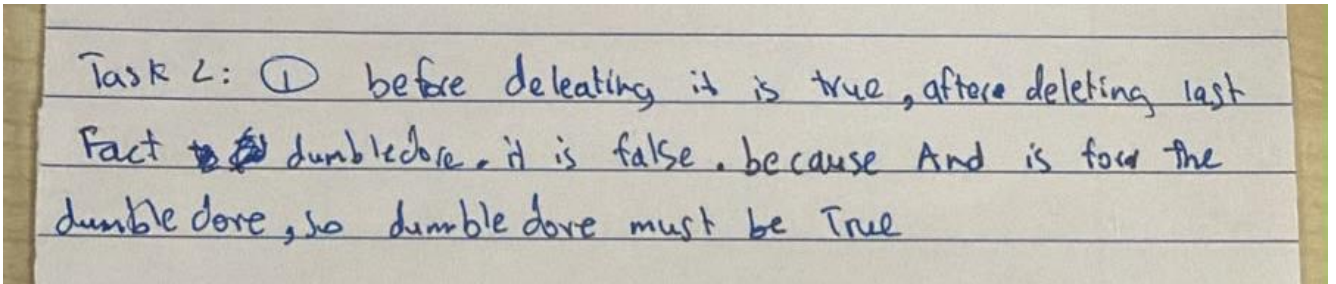
2. **clue.py:** The program prints MsScarlet, library and knife. Explain in plain English why each one must be true.

Answer for Task 1, Q2: The program is about a game to guess the crime by matching the suspect to the weapon used along with the setting where the crime took place. The code used logic operation by using and operation, it only prints true if all the symbols are true. Ms. Scarlet is true, library is true, and knife is true. So, the program prints all the symbols and logically we can guess that the suspect who committed the crime was Ms. Scarlet, the setting is in the library, and the weapon was the knife.

5. Task 2 — Change the Knowledge (30 marks, 20 min)

For each change: (a) predict the result, (b) run the code, (c) explain the result.

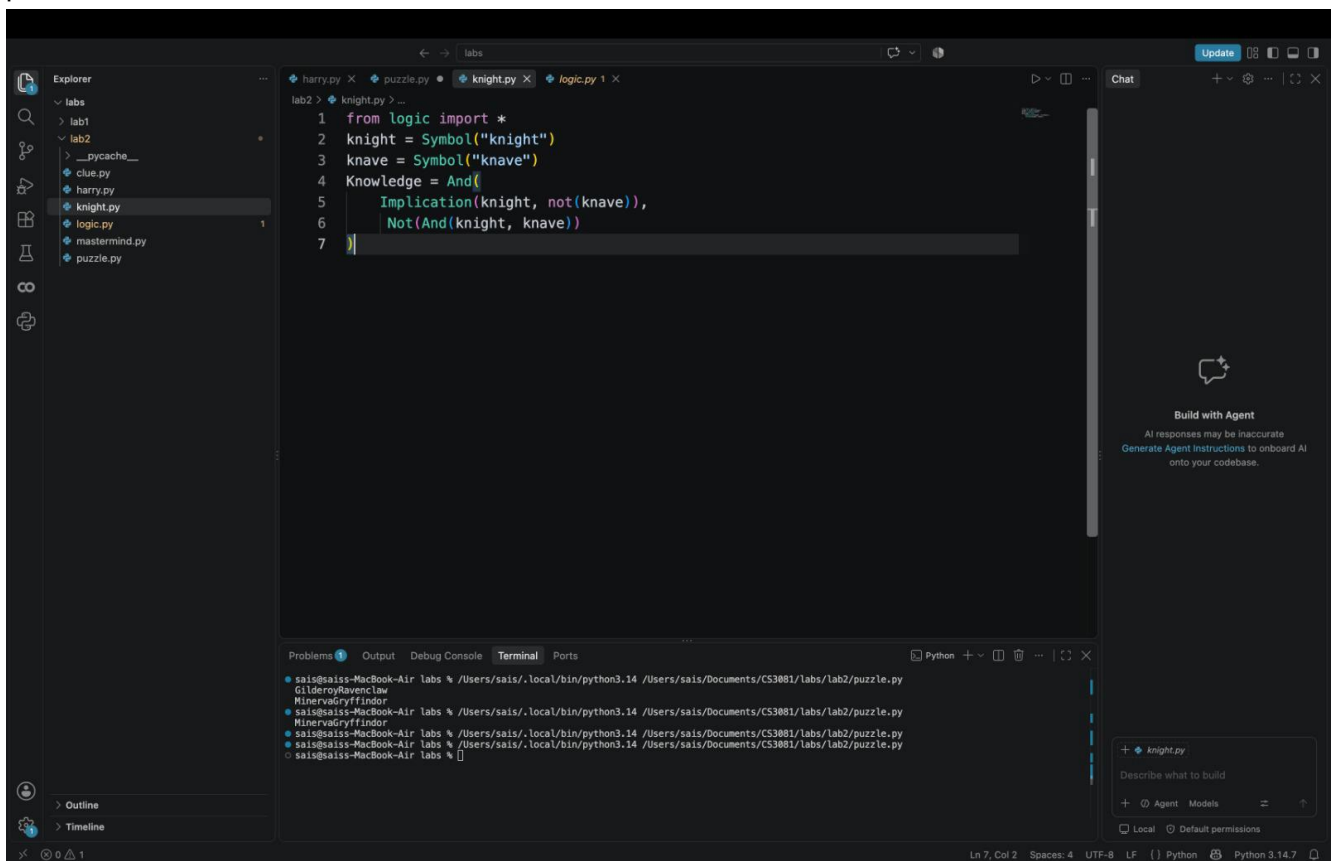
1. **harry.py:** Delete the last fact `dumbledore` from the knowledge base. What does the program print now? Why?



2. **puzzle.py:** Delete the clue `MinervaGryffindor`. Who can still be placed in a house? Then add one new clue of your own that makes the answer unique again.

6. Task 3 — Group Challenge: Knights & Knaves (30 marks, 30 min)

On an island, a **knight** always tells the truth and a **knave** always lies. Every person is one or the other, never both. Create a new file `knights.py` that imports `logic.py` and solves the puzzles below.



Hint — use this pattern for what a person says:

```
Implication(AKnight, statement)      # a knight's statement is true
Implication(AKnave, Not(statement))  # a knave's statement is false
```

1. **Puzzle 0:** A says “I am both a knight and a knave.”
2. **Puzzle 1:** A says “We are both knaves.” B says nothing.
3. **Puzzle 2:** A says “We are the same kind.” B says “We are of different kinds.”

Bonus (+10 marks) — Puzzle 3: A says either “I am a knight.” or “I am a knave.” (you don’t know which). B says “A said ‘I am a knave’.” B also says “C is a knave.” C says “A is a

7. Submission

Create **one public GitHub repository** named `Lab3_GroupX` and submit **only its link** on Blackboard. The repository must contain:

1. `Lab3_GroupX.pdf` — group number, member names, and for each task: your answer, your code, and a screenshot of the output.
2. All `.py` files — `logic.py`, your changed `harry.py` and `puzzle.py`, and your new `knights.py`.
3. `README.md` — member names and one line on how to run `knights.py`.

Before you submit: open your link in a private (incognito) browser window to check that it works.

8. Grading Rubric

Task	What is checked	Marks
Task 1	Correct truth table and clear explanations	30
Task 2	Predictions, results and explanations; a working new clue	30
Task 3	Puzzles 0-2 encoded correctly with output and explanation	30
Submission	Clear PDF and complete GitHub repository	10
Total		100
Bonus	Puzzle 3 solved and explained	+10